

HWRS 561b: Physical Hydrogeology II

Steady-state 1D unsat flow (Part 1)

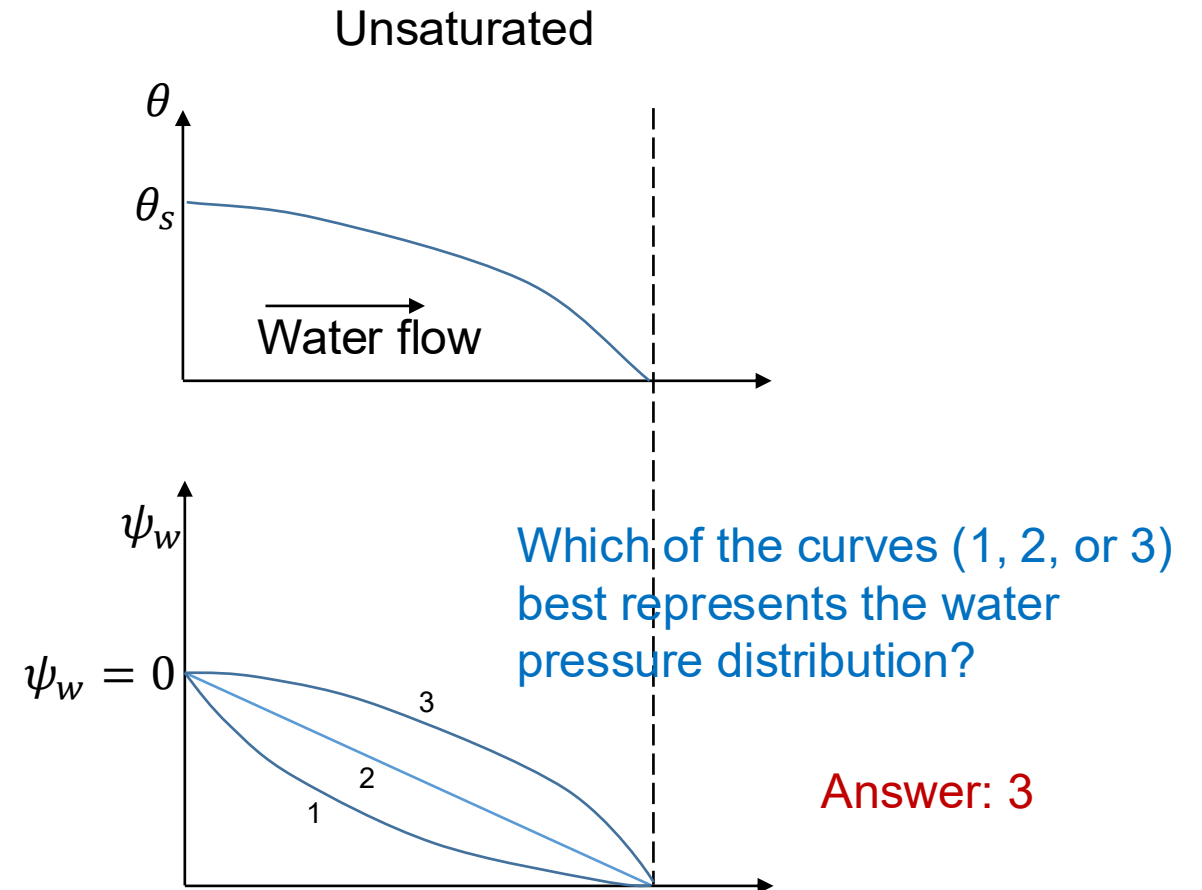
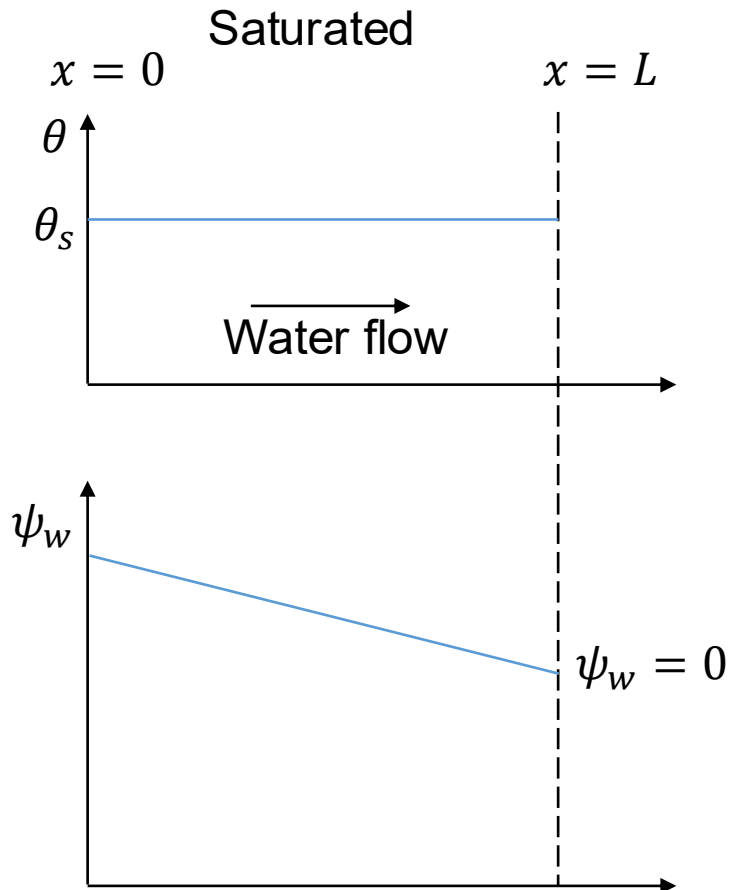
Agenda: Steady-state unsaturated flow
Reading: Chapter 11 (Pinder & Celia, 2006)

- ❖ Three forms of Richards' equation
 - Mixed form
 - Pressure head-based form
 - ✓ Specific moisture capacity
 - Water content-based form
 - ✓ Soil moisture diffusivity (What does the equation have to do with “diffusion”?)
 - ✓ Cannot be used if the domain involves saturated water flow
 - How to include soil and fluid compressibility?
- ❖ Richards' assumptions
 - Air pressure remains almost zero everywhere, but air does move.
 - Does “air movement” make the Richards' equation invalid? No, as long as air pressure remains almost zero everywhere.

Steady-State Unsaturated Flow

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Horizontal flow (Homogeneous column)

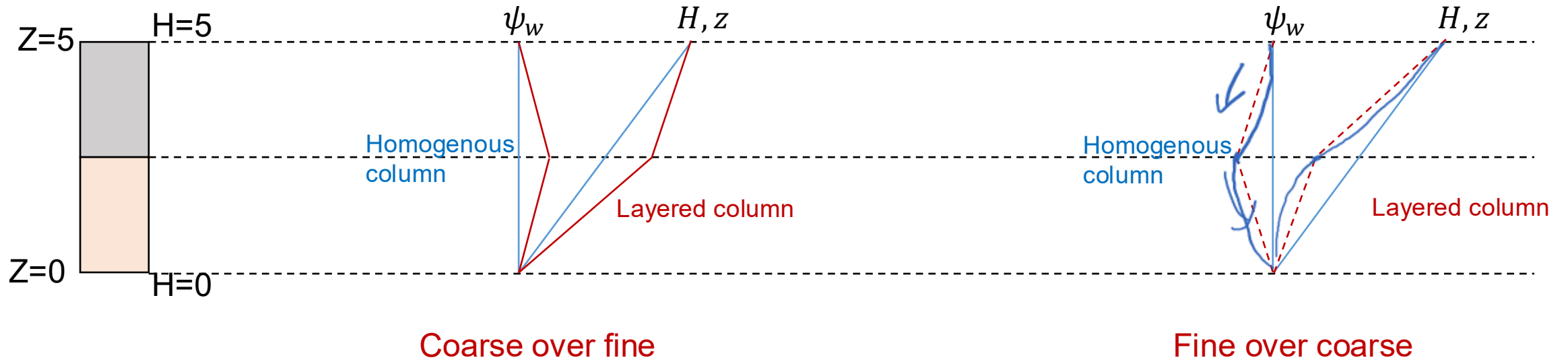


- Unsaturated flow involves nonlinearities that make their behaviors differ from that of the saturated flow

Steady-State Unsaturated Flow

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Vertical flow (in layered columns)

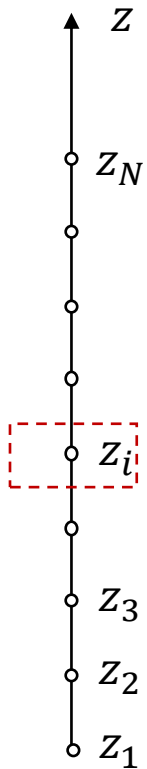


- Hydraulic head and pressure head are both continuous in space.
- Is water saturation continuous?

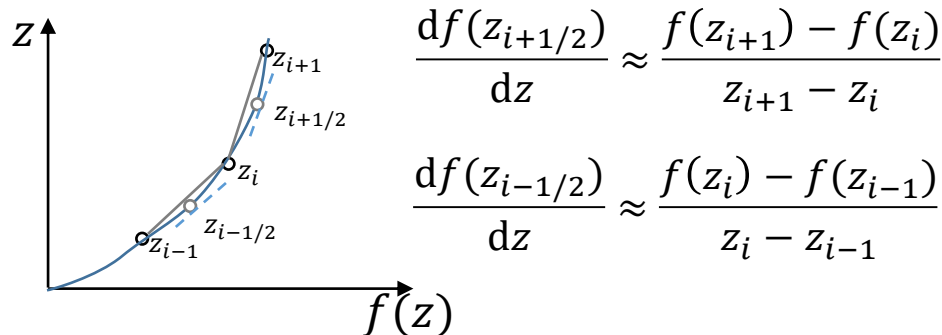
Steady-State Unsaturated Flow: Numerical Soln.

$$\frac{\partial \theta_w}{\partial t} - \frac{\partial}{\partial z} \left(K \frac{\partial \psi_w}{\partial z} \right) - \frac{\partial K}{\partial z} = 0 \quad \xrightarrow{\text{Steady-state}} \quad \frac{d}{dz} \left(K \frac{d\psi_w}{dz} \right) + \frac{dK}{dz} = 0 \quad (1)$$

- Equation (1) is a second-order ordinary differential equation in 1D.
- It is nonlinear because $K = K(\psi_w)$ is a nonlinear function.
- To solve it, we need two boundary conditions and we need to do it in an iterative procedure.

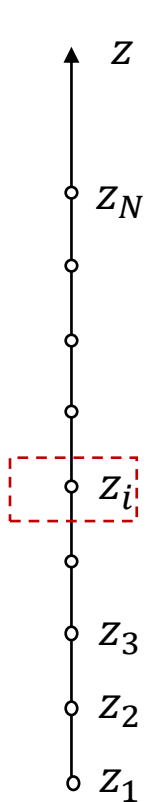


- How to solve this 1D **nonlinear ordinary differential equation**?
- Key idea: Divide the domain into many boxes and convert the **differential equation** to a system of nonlinear **algebraic equations**.
- Technique: Use finite difference to approximate derivatives



Steady-State Unsaturated Flow: Numerical Soln.

$$\cancel{\frac{\partial \theta_w}{\partial t}} - \frac{\partial}{\partial z} \left(K \frac{\partial \psi_w}{\partial z} \right) - \frac{\partial K}{\partial z} = 0 \quad \xrightarrow{\text{Steady-state}} \quad \textcircled{1} \frac{d}{dz} \left(K \frac{d\psi_w}{dz} \right) + \textcircled{2} \frac{dK}{dz} = 0 \quad (1)$$



$$\begin{aligned} \textcircled{1} \frac{d}{dz} \left(K \frac{d\psi_w}{dz} \right) \Big|_{z_i} &\approx \frac{\left(K \frac{d\psi_w}{dz} \right)_{i+1/2} - \left(K \frac{d\psi_w}{dz} \right)_{i-1/2}}{\Delta z} \approx \frac{K_{i+1/2} \frac{\psi_{w,i+1} - \psi_{w,i}}{\Delta z} - K_{i-1/2} \frac{\psi_{w,i} - \psi_{w,i-1}}{\Delta z}}{\Delta z} \\ &= \frac{K_{i+1/2}(\psi_{w,i+1} - \psi_{w,i}) - K_{i-1/2}(\psi_{w,i} - \psi_{w,i-1})}{\Delta z^2} \\ \textcircled{2} \frac{dK}{dz} \Big|_{z_i} &\approx \frac{K_{i+1/2} - K_{i-1/2}}{\Delta z} \\ \Rightarrow \frac{K_{i+1/2}}{\Delta z^2} (\psi_{w,i+1} - \psi_{w,i}) - \frac{K_{i-1/2}}{\Delta z^2} (\psi_{w,i} - \psi_{w,i-1}) + K_{i+1/2} \frac{1}{\Delta z} - K_{i-1/2} \frac{1}{\Delta z} &= 0 \\ \Rightarrow \frac{K_{i+1/2}}{\Delta z^2} \psi_{w,i+1} - \left(\frac{K_{i+1/2}}{\Delta z^2} + \frac{K_{i-1/2}}{\Delta z^2} \right) \psi_{w,i} + \frac{K_{i-1/2}}{\Delta z^2} \psi_{w,i-1} + K_{i+1/2} \frac{1}{\Delta z} - K_{i-1/2} \frac{1}{\Delta z} &= 0 \end{aligned}$$

This is an algebraic equation with 3 unknowns. We can write such an algebraic equation for each node or box and we can get N algebraic equations for the N unknowns $(\psi_{w,1}, \psi_{w,2}, \dots, \psi_{w,N})$

Steady-State Unsaturated Flow: Numerical Soln.

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In matrix form:

$$\begin{pmatrix}
 1 & 0 & 0 & & & & \\
 \frac{K_{3/2}}{\Delta z^2} & -\frac{(K_{3/2} + K_{5/2})}{\Delta z^2} & \frac{K_{5/2}}{\Delta z^2} & & & & \\
 & \frac{K_{5/2}}{\Delta z^2} & -\frac{(K_{5/2} + K_{7/2})}{\Delta z^2} & \frac{K_{7/2}}{\Delta z^2} & & & \\
 & & \ddots & \ddots & \ddots & & \\
 & & & \frac{K_{i-1/2}}{\Delta z^2} & -\frac{(K_{i-1/2} + K_{i+1/2})}{\Delta z^2} & \frac{K_{i+1/2}}{\Delta z^2} & \\
 & & & & \ddots & \ddots & \\
 & & & & & \frac{K_{n-1-1/2}}{\Delta z^2} & -\frac{(K_{n-1-1/2} + K_{n-1+1/2})}{\Delta z^2} & \frac{K_{n-1+1/2}}{\Delta z^2} \\
 & & & & & 0 & 0 & 1
 \end{pmatrix}
 \begin{pmatrix}
 \psi_{w,1} \\
 \psi_{w,2} \\
 \psi_{w,3} \\
 \vdots \\
 \psi_{w,i} \\
 \vdots \\
 \psi_{w,n-1} \\
 \psi_{w,n}
 \end{pmatrix}
 =
 \begin{pmatrix}
 B.C. \\
 \frac{K_{3/2} - K_{5/2}}{\Delta z} \\
 \frac{K_{5/2} - K_{7/2}}{\Delta z} \\
 \vdots \\
 \frac{K_{i-1/2} - K_{i+1/2}}{\Delta z} \\
 \vdots \\
 \frac{K_{n-1-1/2} - K_{n-1+1/2}}{\Delta z} \\
 B.C.
 \end{pmatrix}$$

$\Rightarrow \mathbf{F}(\boldsymbol{\psi}_w) \cdot \boldsymbol{\psi}_w = \mathbf{R}$ How to solve this nonlinear system of algebraic equations?
 $\Rightarrow$ Employ an iterative methods

Steady-State Unsaturated Flow: Numerical Soln.

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$$\frac{\partial \theta_w}{\partial t} + \frac{\partial}{\partial z} \left(-K \left(\frac{\partial \psi_w}{\partial z} + 1 \right) \right) = 0$$

$$q_z = -K \left(\frac{\partial \psi_w}{\partial z} + 1 \right)$$

Assumes z positive upward

$$\frac{\partial \theta_w}{\partial t} + \frac{\partial}{\partial z} \left(-K \left(\frac{\partial \psi_w}{\partial z} - 1 \right) \right) = 0$$

$$q_z = -K \left(\frac{\partial \psi_w}{\partial z} - 1 \right)$$

Assumes z positive downward

Which equation is correct?

Now, we assume z is positive upward.

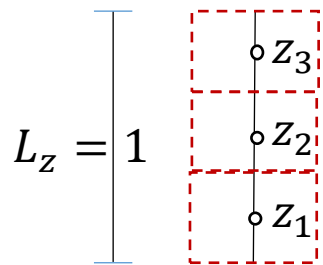
Which equation is correct?

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$$\frac{d}{dz} \left(K \frac{d\psi_w}{dz} \right) + \frac{dK}{dz} = 0$$

$$\psi_w(z_3) = 2$$



$$\psi_w(z_1) = 0$$

$$\Delta z = z_2 - z_1 = z_3 - z_2 = 1/3$$

Equation for box ①:

$$\psi_{w,1} = 0$$

Equation for box ③:

$$\psi_{w,3} = 2$$

Equation for box ②:

$$\frac{\left(K \frac{d\psi_w}{dz} \right) \Big|_{z+\frac{\Delta z}{2}} - \left(K \frac{d\psi_w}{dz} \right) \Big|_{z-\frac{\Delta z}{2}}}{\Delta z} + \frac{K_{z+\frac{\Delta z}{2}} - K_{z-\frac{\Delta z}{2}}}{\Delta z} = 0$$

$$\frac{K_{z+\frac{\Delta z}{2}} \frac{\psi_{w,3} - \psi_{w,2}}{\Delta z} - K_{z-\frac{\Delta z}{2}} \frac{\psi_{w,2} - \psi_{w,1}}{\Delta z}}{\Delta z} + \frac{K_{z+\frac{\Delta z}{2}} - K_{z-\frac{\Delta z}{2}}}{\Delta z} = 0$$

$$\Rightarrow \frac{K_{z+\frac{\Delta z}{2}}}{\Delta z^2} \psi_{w,3} + \left(-\frac{K_{z+\frac{\Delta z}{2}}}{\Delta z^2} + \frac{K_{z-\frac{\Delta z}{2}}}{\Delta z^2} \right) \psi_{w,2} + \frac{K_{z-\frac{\Delta z}{2}}}{\Delta z^2} \psi_{w,1} = -\frac{K_{z+\frac{\Delta z}{2}} - K_{z-\frac{\Delta z}{2}}}{\Delta z}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{K_{z+\frac{\Delta z}{2}}}{\Delta z^2} & -\frac{K_{z+\frac{\Delta z}{2}}}{\Delta z^2} + \frac{K_{z-\frac{\Delta z}{2}}}{\Delta z^2} & \frac{K_{z+\frac{\Delta z}{2}}}{\Delta z^2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \psi_{w,1} \\ \psi_{w,2} \\ \psi_{w,3} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{K_{z+\frac{\Delta z}{2}} - K_{z-\frac{\Delta z}{2}}}{\Delta z} \\ 2 \end{bmatrix}$$

For our given B.C.:

$$\psi_{w,2} = 1$$

⇒ Everywhere is saturated

For another B.C.:

$$\text{e.g.: } \psi_{w,1} = -3$$

$$\psi_{w,3} = -1$$

need to solve iteratively